

Quiz 8

Friday, November 6 — products, powers, and computational cost (Week 10)

Last updated: October 8, 2026.

Covers the week before only: [Products](#), [Powers](#), [Cost](#), and the [micro-lab](#) (Week 10). Same-day [determinants](#) and this week's inverses/rank lectures are **not** on this quiz.

Textbooks: [Boyd & Vandenberghe \(VMLS\)](#) · also on [Resources](#).

[Schedule](#)

Core

Fluency (complete prompts — practice these by hand)

C8.1. Let $a = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, and $c = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$.

- (a) Form the outer product ab^T (give the 3×3 matrix).
- (b) Compute $(ab^T)c$ and $a(b^Tc)$. Confirm they are equal as vectors.

C8.2. In C8.1, which evaluation order stores a full 3×3 intermediate matrix? Which only needs a scalar intermediate b^Tc ? In one sentence, why does the second order win for large n when $a, b, c \in \mathbb{R}^n$?

C8.3. Let

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Interpret ABx as “apply B , then A .” Compute Bx , then $A(Bx)$. Also compute BAx . Is $AB = BA$?

C8.4. Dimensions. For each pair, decide whether AB is defined and give the size of AB if it is.

- (a) A is 4×2 , B is 2×3 .
- (b) A is 3×2 , B is 3×2 .
- (c) A is 2×2 , B is 2×5 .

C8.5. Compute

$$\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}$$

by forming columns Ab_1 and Ab_2 (or by row-column inner products).

C8.6. Stage 1: $B = \text{diag}(2, 1)$. Stage 2: $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (90° CCW). Write the single matrix M for “ B then A ” and compute $M(1, 1)$.

C8.7. Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (swap coordinates) and $x_0 = (4, 1)$. Compute A^2 and $x_2 = A^2x_0$. Interpret A^2 in one sentence.

C8.8. Directed graph on nodes $\{1, 2, 3\}$ with adjacency (Boyd: $A_{ij} = 1$ means edge from j to i)

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Given

$$A^2 = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix},$$

what does $(A^2)_{31}$ mean? How many length-2 walks from node 1 to node 3?

C8.9. For dynamics $x_{t+1} = Ax_t$, true or false with a one-line reason: $x_5 = A^5x_0$ and also $x_5 = Ax_4$.

C8.10 (micro-lab / cost). You need $y = (ab^\top)c$ for tall vectors of length $n = 1000$. Roughly how many numbers are stored in the intermediate $ab^\top$? Roughly how many in the scalar $b^\top c$? Which plan would you time in a micro-lab, and why?

Textbook Core

- B 10.3, 10.5, 10.6, 10.7, 10.9, 10.10, 10.13, 10.14, 10.15, 10.22

Stretch

B 10.18, 10.23.

What a quiz item might change

The order of two transformations, matrix dimensions, vectors in the $ab^\top c$ comparison, or timing output from two equivalent computations.